

REPORT
OF
Data Analytics Internship
at CodeAlpha

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B. Tech CSE



By

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CANDIDATE'S DECLARATION

I DEEPTI GARKOTI, do hereby declare that the internship report titled "Data Analytics Internship at CodeAlpha" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to CodeAlpha for providing me with the opportunity to undertake the Data Analytics Internship. This internship gave me practical exposure to data analytics concepts and helped me strengthen my knowledge of Python, data analysis, web scraping, and data visualization through hands-on projects.

I am deeply thankful to the School of Computer Science and Engineering, IILM University, for encouraging practical learning and providing continuous academic guidance throughout the internship period.

I also extend my heartfelt thanks to my faculty members for their valuable support, motivation, and guidance. Finally, I express my sincere gratitude to my parents and family for their constant encouragement and support, which motivated me to successfully complete this internship and prepare this report.

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INTERNSHIP COMPLETION CERTIFICATE



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4. PROJECT DESCRIPTION

4.1 Introduction

During my internship at CodeAlpha, I got the opportunity to work on practical Data Analytics tasks using Python and Jupyter Notebook. The internship helped me understand how data is handled in a real project, starting from collecting and preparing the data to analyzing it and presenting the results through visualizations.

As part of the internship, I worked on three main tasks. The first task was Web Scraping, where I collected book-related information from the Books to Scrape website using Python. I used libraries such as Requests and BeautifulSoup to collect and extract the required information and then organized the data using Pandas.

The second task was Exploratory Data Analysis (EDA). For this task, I used an Amazon sales dataset downloaded from Kaggle. I imported the dataset into Jupyter Notebook and examined its structure, columns, and values. I also performed basic data cleaning and explored the dataset to understand different patterns in the data.

The third task was Data Visualization. I used Matplotlib and Seaborn to create different graphs and charts from the dataset. These visualizations made it easier to understand the data and identify useful patterns related to the products, sales, and other attributes.

Working on these tasks gave me practical experience with Python and different data analytics libraries. More importantly, I learned how the different stages of data analytics collecting, cleaning, exploring, and visualizing data work together to turn raw data into useful information.

4.2 Organization Profile

CodeAlpha is an Ed-Tech platform that focuses on practical learning, virtual internships, project-based training, and career development for students and beginners in the technology field. The organization provides learners with opportunities to gain practical exposure and develop technical skills through hands-on assignments and projects.

CodeAlpha offers internship programs in various technology domains, including Python, Web Development, App Development, Artificial Intelligence and Machine Learning, Data Analysis, and UI/UX Design. These internships are designed to help students apply their academic knowledge to practical tasks and become familiar with tools and technologies used in their respective fields.

The organization follows a project-based approach to learning. Instead of focusing only on theoretical concepts, interns are given practical assignments that require them to work with real datasets, programming tools, and different technologies. This approach helps students improve their technical knowledge, problem-solving ability, and confidence in handling practical work.

As part of my internship, I worked in the Data Analytics domain. My assigned work included Web Scraping, Exploratory Data Analysis, and Data Visualization using Python and libraries such as Pandas, NumPy, Matplotlib, Seaborn, Requests, and BeautifulSoup. These tasks gave me an opportunity to understand the basic workflow of data analytics, from collecting and preparing data to analyzing and presenting it through visualizations.

CodeAlpha also provides internship completion certificates after successful completion of the required tasks. Such practical experience and certification can be useful for students while building their academic and professional portfolios.

Overall, CodeAlpha provided me with an opportunity to gain practical experience in Data Analytics and apply concepts learned during my academic studies to project-based tasks.

4.3 Problem Statement

During the internship, the main challenge was to work with data in its different forms and understand how it can be converted into useful information. Before analysis, the data needed to be collected, organized, and checked properly so that meaningful observations could be made.

For the web scraping task, the challenge was to collect the required information from a website and convert the unstructured webpage content into a structured dataset. This required understanding how webpages are organized and using Python libraries to extract the relevant information.

For the Exploratory Data Analysis task, the main challenge was to work with an Amazon sales dataset and understand its structure, values, and patterns. The dataset needed to be examined and cleaned before performing the analysis so that the results would be more reliable and easier to interpret.

Another challenge was presenting the analyzed information in a simple and understandable way. Creating suitable graphs and choosing the right type of visualization was important for identifying patterns and communicating the findings clearly.

Therefore, the overall problem addressed during the internship was to collect, prepare, analyse, and visualize data using Python-based tools and techniques.

4.4 Project Objectives

The main objective of this internship was to gain practical experience in Data Analytics and understand how data can be collected, prepared, analyzed, and presented using Python. I wanted to apply the concepts I had learned during my studies to practical tasks and become more confident in working with datasets.

The specific objectives of the internship were:

1. To improve my understanding of Python and its use in practical Data Analytics tasks.
2. To learn how to collect data from websites using web scraping techniques and libraries such as Requests and BeautifulSoup.
3. To understand data cleaning and preprocessing and learn why these steps are important before analyzing a dataset.
4. To gain practical experience with Pandas and NumPy for handling, manipulating, and analyzing data.
5. To perform Exploratory Data Analysis (EDA) and identify useful patterns and relationships within a dataset.
6. To create meaningful visualizations using Matplotlib and Seaborn to present data in an easy-to-understand form.
7. To become more comfortable working in Jupyter Notebook and organizing Python-based analysis.
8. To understand the overall Data Analytics workflow from collecting raw data to obtaining useful insights.
9. To improve my problem-solving and analytical skills by working on practical internship tasks.
10. To gain confidence in applying Data Analytics concepts to real-world datasets and practical situations.

4.5 Scope of the Project

The scope of this internship was to gain practical experience in the basic stages of Data Analytics by working with datasets and Python-based tools. The project mainly focused on collecting data, preparing it for analysis, exploring the available information, and presenting the findings through suitable visualizations.

The first part of the work involved web scraping, where information was collected from the Books to Scrape website. This helped me understand how data available on webpages can be extracted and converted into a structured format for further use.

The second part focused on Exploratory Data Analysis using an Amazon sales dataset obtained from Kaggle. The scope of this work included loading the dataset, understanding its structure, checking the available information, performing basic data cleaning, and exploring patterns within the data.

The project also included data visualization, where graphs and charts were created using Matplotlib and Seaborn. These visualizations helped in presenting the information in a clearer form and made it easier to identify patterns and relationships in the dataset.

Through these activities, the scope of the internship extended beyond simply writing Python code. It helped me understand how different tools and techniques can be combined to work with data from its initial collection to its final presentation.

Overall, the project provided me with a practical foundation in Data Analytics and helped me understand how the skills learned during the internship can be applied to datasets in a structured manner.

4.6 Technologies and Tools Used

During my internship, I used different Python libraries and development tools depending on the requirements of each task. These tools helped me collect, organize, analyse, and visualize the data. The main technologies and tools used during the internship are listed below.

Technology / Tool	Purpose
Python	Main programming language used for performing Data Analytics tasks.
Jupyter Notebook	Used for writing, running, and testing Python code and documenting the analysis.
Pandas	Used for loading, cleaning, organizing, and analyzing datasets.
NumPy	Used for numerical operations and handling data efficiently.
Matplotlib	Used to create basic charts and graphs for data analysis.
Seaborn	Used to create statistical and informative visualizations.
Requests	Used to send HTTP requests and retrieve webpage content during web scraping.
BeautifulSoup	Used to parse HTML content and extract required information from webpages.
Kaggle	Used as the source for the Amazon sales dataset used during analysis.

These tools worked together throughout the internship. Python provided the programming environment, while libraries such as Pandas and NumPy helped with data handling and analysis. Requests and BeautifulSoup were useful for collecting web data, and Matplotlib and Seaborn helped me present the analyzed data through visualizations.

4.7 System Architecture

The overall workflow followed during the internship can be represented as a simple data analytics process. The work started with collecting the required data, followed by cleaning and organizing the data. After that, the data was explored to understand its structure and identify useful patterns. Finally, visualizations were created to present the findings in a clear and understandable manner.



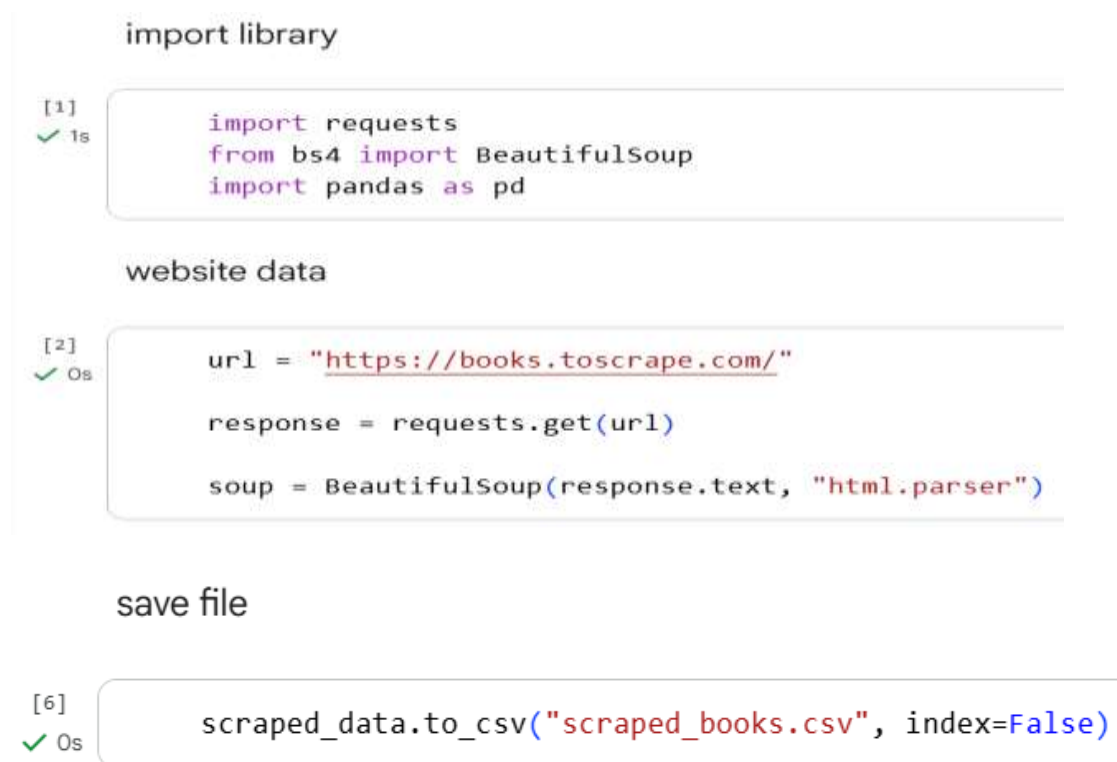
Figure 4.1: Overall Data Analytics Workflow

4.8 METHODOLOGY

The methodology followed during my internship was divided into three main tasks: Web Scraping, Exploratory Data Analysis, and Data Visualization. Each task helped me understand a different stage of the Data Analytics process. I completed the tasks using Python and different libraries according to the requirements of the work.

Task 1: Web Scraping

The first task was to collect data from the Books to Scrape website. I used the Requests library to access the webpage and BeautifulSoup to extract the required information from the HTML content. The collected book details were then organized into a structured format using Pandas and saved as a dataset for further use.



```
import library

[1] ✓ 1s
import requests
from bs4 import BeautifulSoup
import pandas as pd

website data

[2] ✓ 0s
url = "https://books.toscrape.com/"
response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")

save file

[6] ✓ 0s
scraped_data.to_csv("scraped_books.csv", index=False)
```

Figure 4.2: Python code used for Web Scraping

The code retrieves the webpage, parses its HTML content, and prepares the required information for further processing.

Task 2: Exploratory Data Analysis

The second task involved analyzing an Amazon sales dataset downloaded from Kaggle. I first loaded the dataset into Jupyter Notebook using Pandas and checked its structure, columns, and values. I then performed basic data cleaning and explored the dataset to understand the available information and identify useful patterns.



Figure 4.3: Python code used for Exploratory Data Analysis

The code imports the required Python libraries, loads the Amazon sales dataset, and displays the initial data for further exploration and analysis.

Task 3: Data Visualization

In the third task, I used the analyzed data to create different charts and graphs. Matplotlib and Seaborn were used to present the information visually. The visualizations helped me understand the data more easily and made it simpler to communicate important observations.

The three tasks followed a connected workflow. Data was first collected or obtained, then prepared and explored, and finally presented through visualizations. This approach helped me understand how raw data can be gradually converted into meaningful information.

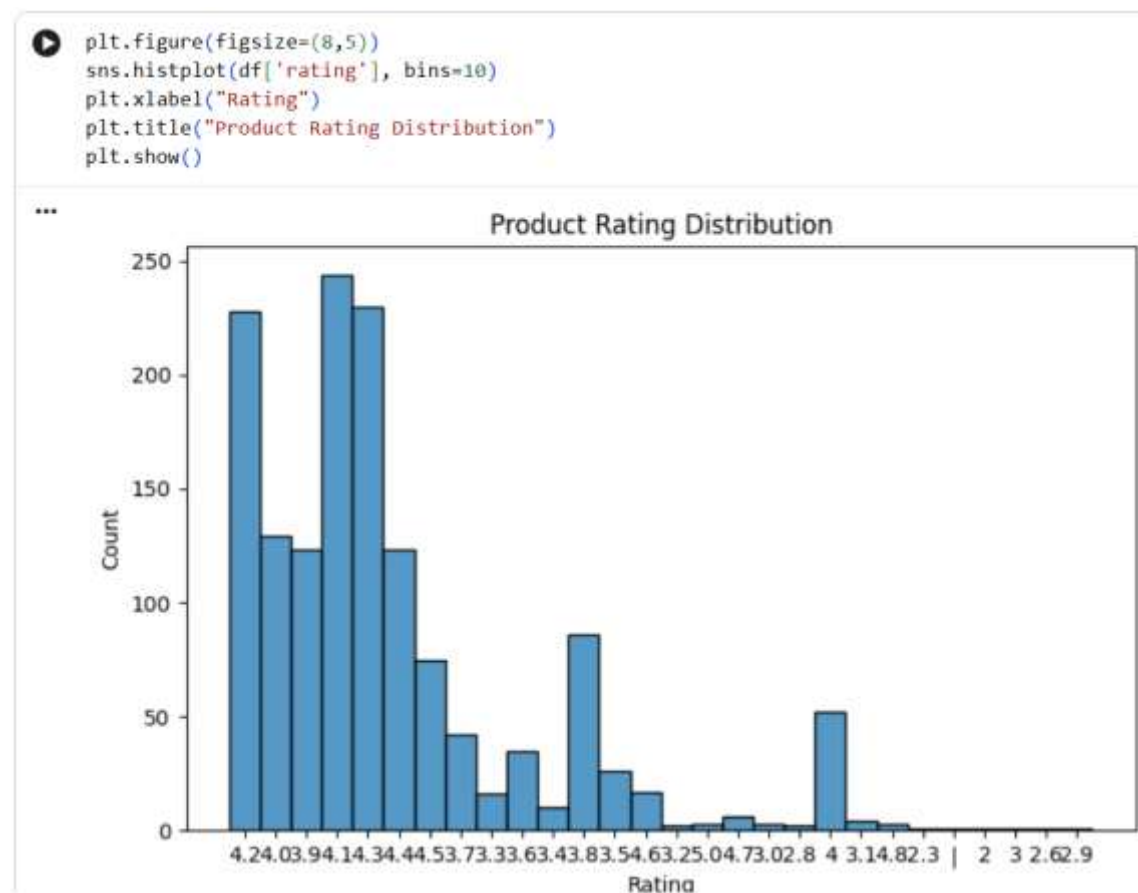


Figure 4.4: Python code used for Data Visualization

The visualization shows the distribution of product ratings in the dataset. The histogram indicates that most products have ratings concentrated in the higher rating range, while fewer products have very low or very high ratings. This visualization makes it easier to understand the overall distribution of product ratings.

4.9 EXPECTED OUTCOMES

The internship was expected to help me develop a better understanding of the practical use of Data Analytics. By working on the assigned tasks, I aimed to improve both my technical skills and my ability to work with data in a structured manner.

One of the expected outcomes was to become more comfortable with Python and commonly used libraries such as Pandas, NumPy, Matplotlib, Seaborn, Requests, and BeautifulSoup. I also expected to understand how these tools can be used together for different stages of data analysis.

Through the Web Scraping task, I expected to learn how information can be collected from webpages and converted into a structured dataset. The Exploratory Data Analysis task was expected to improve my understanding of data cleaning, data exploration, and identifying patterns within a dataset.

The Data Visualization task was expected to help me understand how graphs and charts can be used to present data in a clear and meaningful way. I also wanted to improve my ability to interpret visual patterns and communicate observations from the data.

Overall, I expected the internship to give me practical exposure to the complete basic workflow of Data Analytics and help me become more confident in applying the concepts I had learned during my academic studies.

The major expected outcomes were:

1. Better understanding of practical Data Analytics.
2. Improved Python programming and data-handling skills.
3. Practical experience with data collection and web scraping.
4. Better understanding of data cleaning and EDA.
5. Improved ability to create and interpret visualizations.
6. Greater confidence in working with real datasets.

5. BIBLIOGRAPHY / REFERENCES

The following resources were referred to during the internship for learning, data collection, analysis, and understanding the tools used in the project.

1. **CodeAlpha** — Internship tasks, instructions, and internship-related resources.
2. **Books to Scrape** — Website used for the Web Scraping task.
3. **Kaggle** — Source of the Amazon sales dataset used for Exploratory Data Analysis.
4. **Python Documentation** — Reference for Python programming concepts and libraries.
5. **Pandas Documentation** — Reference for data manipulation and analysis.
6. **NumPy Documentation** — Reference for numerical operations.
7. **Matplotlib Documentation** — Reference for data visualization.
8. **Seaborn Documentation** — Reference for statistical data visualization.
9. **BeautifulSoup Documentation** — Reference for HTML parsing and web scraping.
10. **Requests Documentation** — Reference for making HTTP requests.